Proposed Prescribing Information For the use of a Registered Medicinal Practitioner or a Hospital or a Laboratory only

EXCARB TABLETS 200mg/400mg/600mg /800 mg
(Eslicarbazepine Acetate Tablets 200mg/400mg/600mg/800mg)

1. COMPOSITION

Eslicarbazepine Acetate Tablets 200mg
Each uncoated tablet contains:
Eslicarbazepine Acetate................200 mg
Excipients .... q.s.

Eslicarbazepine Acetate Tablets 400mg
Each uncoated tablet contains:
Eslicarbazepine Acetate................400 mg
Excipients .... q.s.

Eslicarbazepine Acetate Tablets 600mg
Each uncoated tablet contains:
Eslicarbazepine Acetate................600 mg
Excipients .... q.s.

Eslicarbazepine Acetate Tablets 800mg
Each uncoated tablet contains:
Eslicarbazepine Acetate................800 mg
Excipients .... q.s.

2. PHARMACEUTICAL FORM

Uncoated Tablet.

Eslicarbazepine Acetate Tablets 200 mg:
White to off-white, oblong, uncoated tablets debossed with 'V1' on one side and break line on the other side.

Eslicarbazepine Acetate Tablets 400 mg:
White to off-white, circular, biconvex, uncoated tablets debossed with 'V2' on one side and plain on the other side.

Eslicarbazepine Acetate Tablets 600 mg:
White to off-white, oblong, uncoated tablets debossed with 'V3' on one side and break line on the other side.

Eslicarbazepine Acetate Tablets 800 mg:
White to off-white, oblong, uncoaed tablets debossed with 'V7' on one side and break line on the other side.

3. CLINICAL PARTICULARS

3.1 Indication

Eslicarbazepine Acetate tablet is indicated as:
- monotherapy in the treatment of partial-onset seizures, with or without secondary generalisation, in adults with newly diagnosed epilepsy;

- adjunctive therapy in adults, adolescents and children aged above 6 years, with partial-onset seizures with or without secondary generalisation.

3.2 Posology and method of administration

Posology

Adults
Eslicarbazepine Acetate tablets may be taken as monotherapy or added to existing anticonvulsant therapy. The recommended starting dose is 400 mg once daily which should be increased to 800 mg once daily after one or two weeks. Based on individual response, the dose may be increased to 1,200 mg once daily. Some patients on monotherapy regimen may benefit from a dose of 1,600 mg once daily.

Special populations

Elderly (over 65 years of age)
No dose adjustment is needed in the elderly population provided that the renal function is not disturbed. Due to very limited data on the 1,600 mg monotherapy regimen in the elderly, this dose is not recommended for this population.

Renal impairment
Caution should be exercised in the treatment of patients, adult and children above 6 years of age, with renal impairment and the dose should be adjusted according to creatinine clearance (CL_{CR}) as follows:
- CL_{CR} >60 ml/min: no dose adjustment required.
- CL_{CR} 30-60 ml/min: initial dose of 200 mg (or 5 mg/kg in children above 6 years) once daily or 400 mg (or 10 mg/kg in children above 6 years) every other day for 2 weeks followed by a once daily dose of 400 mg (or 10 mg/kg in children above 6 years). However, based on individual response, the dose may be increased.
- CL_{CR} <30 ml/min: use is not recommended in patients with severe renal impairment due to insufficient data.

Hepatic impairment
No dose adjustment is needed in patients with mild to moderate hepatic impairment.
The pharmacokinetics of eslicarbazepine acetate has not been evaluated in patients with severe hepatic impairment and use in these patients is, therefore, not recommended.

Paediatric population
Children above 6 years of age
The recommended starting dose is 10 mg/kg/day once daily. Dosage should be increased in weekly or bi-weekly increments of 10 mg/kg/day up to 30 mg/kg/day, based on individual response. The maximum dose is 1,200 mg once daily.

Children with a body weight of ≥60 kg
Children with a body weight of 60 kg or more should be given the same dose as for adults.
The safety and efficacy of eslicarbazepine acetate in children aged 6 years and below has not yet been established.

Method of administration

Oral use.
Eslicarbazepine Acetate tablets may be taken with or without food.
The tablet can be split, except for the 400 mg tablet. Do not crush or mix the tablet with any other liquids.

3.3 Contraindications

Hypersensitivity to the active substance, to other carboxamide derivatives (e.g. carbamazepine, oxcarbazepine) or to any of the excipients used in the formulation.

Second or third degree atrioventricular (AV) block.

3.4 Special warnings and precautions for use

Suicidal ideation

Suicidal ideation and behaviour have been reported in patients treated with antiepileptic active substances in several indications. A meta-analysis of randomised placebo-controlled trials of antiepileptic medicinal products has also shown a small increased risk of suicidal ideation and behaviour. The mechanism of this risk is not known, and the available data do not exclude the possibility of an increased risk for eslicarbazepine acetate. Therefore, patients should be monitored for signs of suicidal ideation and behaviours and appropriate treatment should be considered. Patients (and caregivers of patients) should be advised to seek medical advice should signs of suicidal ideation or behaviour emerge.

Nervous System disorders

Eslicarbazepine acetate has been associated with some central nervous system adverse reactions, such as dizziness and somnolence, which could increase the occurrence of accidental injury.

Other warnings and precautions

If eslicarbazepine acetate tablet is to be discontinued, it is recommended to withdraw it gradually to minimise the potential of increased seizure frequency.

Cutaneous reactions

Rash developed as an adverse reaction in 1.2% of total population treated with eslicarbazepine in clinical studies in epileptic patients. Urticaria and angioedema cases have been reported in patients taking eslicarbazepine. Angioedema in the context of hypersensitivity/anaphylactic reaction associated with laryngeal oedema can be fatal. If signs or symptoms of hypersensitivity develop, eslicarbazepine acetate must be discontinued immediately and alternative treatment should be initiated.
Severe cutaneous adverse reactions (SCARS) including Stevens-Johnson syndrome (SJS)/toxic epidermal necrolysis (TEN) and drug reaction with eosinophilia and systemic symptoms (DRESS), which can be life-threatening or fatal, have been reported in post-marketing experience with eslicarbazepine treatment. At the time of prescription patients should be advised of the signs and symptoms and monitored closely for skin reactions. If signs and symptoms suggestive of these reactions appear, eslicarbazepine Acetate tablets should be withdrawn immediately and an alternative treatment considered (as appropriate). If the patients have developed such reactions, treatment with eslicarbazepine Acetate tablets must not be restarted in these patients at any time.

HLA-B* 1502 allele - in Han Chinese, Thai and other Asian populations

HLA-B* 1502 in individuals of Han Chinese and Thai origin has been shown to be strongly associated with the risk of developing the severe cutaneous reactions known as Stevens Johnson syndrome (SJS) when treated with carbamazepine. The chemical structure of eslicarbazepine acetate is similar to that of carbamazepine, and it is possible that patients who are positive for HLA-B*1502 may also be at risk for SJS after treatment with eslicarbazepine acetate.

The prevalence of HLA-B*1502 carrier is about 10% in Han Chinese and Thai populations. Whenever possible, these individuals should be screened for this allele before starting treatment with carbamazepine or chemically-related active substances.
If patients of these ethnic origins are tested positive for HLA- B*1502 allele, the use of eslicarbazepine acetate may be considered if the benefits are thought to exceed risks.
Because of the prevalence of this allele in other Asian populations (e.g, above 15% in the Philippines and Malaysia), testing genetically at risk populations for the presence of HLA- B*1502 may be considered.

HLA-A*3101 allele- European descent and Japanese populations

There are some data that suggest HLA-A*3101 is associated with an increased risk of carbamazepine induced cutaneous adverse drug reactions including SJS, TEN, Drug rash with eosinophilia (DRESS), or less severe acute generalized exanthematous pustulosis (AGEP) and maculopapular rash in people of European descent and the Japanese.
The frequency of the HLA-A*3101 allele varies widely between ethnic populations. HLA-A*3101 allele has a prevalence of 2 to 5% in European populations and about 10% in Japanese population. The presence of HLA-A*3101 allele may increase the risk for carbamazepine induced cutaneous reactions (mostly less severe) from 5.0% in general population to 26.0% among subjects of European ancestry, whereas its absence may reduce the risk from 5.0% to 3.8%.
There are insufficient data supporting a recommendation for HLA-A*3101 screening before starting carbamazepine or chemically-related compounds treatment.
If patients of European descent or Japanese origin are known to be positive for HLA-A*3101 allele, the use of carbamazepine or chemically-related compounds may be considered if the benefits are thought to exceed risks.

Hyponatraemia

Hyponatraemia has been reported as an adverse reaction in 1.5% of patients treated with eslicarbazepine. Hyponatraemia is asymptomatic in most cases, however, it may be accompanied by clinical symptoms like worsening of seizures, confusion, decreased consciousness. Frequency of hyponatraemia increased with increasing eslicarbazepine acetate dose. In patients with pre-existing renal disease leading to hyponatraemia, or in patients concomitantly treated with medicinal products which may themselves lead to hyponatraemia (e.g. diuretics, desmopressin, carbamazepine), serum sodium levels should be examined before and during treatment with eslicarbazepine acetate. Furthermore, serum sodium levels should be determined if clinical signs of hyponatraemia occur. Apart from this, sodium levels should be determined during routine laboratory examination. If clinically-relevant hyponatraemia develops, eslicarbazepine acetate should be discontinued.

PR interval
Prolongations in PR interval have been observed in clinical studies with eslicarbazepine acetate. Caution should be exercised in patients with medical conditions (e.g. low levels of thyroxine, cardiac conduction abnormalities), or when taking concomitant medicinal products known to be associated with PR prolongation.

Renal impairment
Caution should be exercised in the treatment of patients with renal impairment and the dose should be adjusted according to creatinine clearance. In patients with CLCR <30 ml/min use is not recommended due to insufficient data.

Hepatic impairment
As clinical data are limited in patients with mild to moderate hepatic impairment and pharmacokinetic and clinical data are missing in patients with severe hepatic impairment, eslicarbazepine acetate should be used with caution in patients with mild to moderate hepatic impairment and is not recommended in patients with severe hepatic impairment.

Eslicarbazepine Acetate tablets contains sodium
This medicine contains less than 1mmol sodium (23 mg) per tablet, that is to say essential 'sodium-free'.

3.5 Interaction with other medicinal products and other forms of interaction

Interaction studies have only been performed in adults.

Eslicarbazepine acetate is extensively converted to eslicarbazepine, which is mainly eliminated by glucuronidation. *In vitro* eslicarbazepine is a weak inducer of CYP3A4 and UDP-glucuronyl transferases. *In vivo* eslicarbazepine showed an inducing effect on the metabolism of medicinal products that are mainly eliminated by metabolism through CYP3A4 (e.g. Simvastatin). Thus, an increase in the dose of the medicinal products that are mainly metabolised through CYP3A4 may be required, when used concomitantly with eslicarbazepine acetate. Eslicarbazepine *in vivo* may have an inducing effect on the metabolism of medicinal products that are mainly eliminated by conjugation through the UDP-glucuronyl transferases. When initiating or discontinuing treatment with eslicarbazepine acetate tablets or changing the dose, it may take 2 to 3 weeks to reach the new level of enzyme activity. This time delay must be taken into account when eslicarbazepine acetate tablets is being used just prior to or in combination with other medicinal products that require dose adjustment when co-administered with eslicarbazepine acetate tablets. Eslicarbazepine has inhibiting properties with respect to CYP2C19. Thus, interactions can arise when co-administering high doses of eslicarbazepine acetate with medicinal products that are mainly metabolised by CYP2C19 (e.g. Phenytoin).

Interactions with other antiepileptic medicinal Products

Carbamazepine
In a study in healthy subjects, concomitant administration of eslicarbazepine acetate 800 mg once daily and carbamazepine 400 mg twice daily resulted in an average decrease of 32% in exposure to the active metabolite eslicarbazepine, most likely caused by an induction of glucuronidation. No change in exposure to carbamazepine or its metabolite carbamazepine-epoxide was noted. Based on individual response, the dose of eslicarbazepine acetate may need to be increased if used concomitantly with carbamazepine. Results from patient studies showed that concomitant treatment increased the risk of the following adverse reactions: diplopia, abnormal coordination and dizziness. The risk of increase of other specific adverse reactions caused by co-administration of carbamazepine and eslicarbazepine acetate cannot be excluded.

Phenytoin
In a study in healthy subjects, concomitant administration of eslicarbazepine acetate 1,200 mg once daily and phenytoin resulted in an average decrease of 31-33% in exposure to the active metabolite, eslicarbazepine, most likely caused by an induction of glucuronidation, and an average increase of 31-35% in exposure to phenytoin, most likely caused by an inhibition of CYP2C19. Based on individual response, the dose of eslicarbazepine acetate may need to be increased and the dose of phenytoin may need to be decreased.

Lamotrigine
Glucuronidation is the major metabolic pathway for both eslicarbazepine and lamotrigine and, therefore, an interaction could be expected. A study in healthy subjects with eslicarbazepine acetate 1,200 mg once daily showed a minor average pharmacokinetic interaction (exposure of lamotrigine decreased 15%) between eslicarbazepine acetate and lamotrigine and consequently no dose adjustments are required. However, due to inter-individual variability, the effect may be clinically relevant in some individuals.

Topiramate
In a study in healthy subjects, concomitant administration of eslicarbazepine acetate 1,200 mg once daily and topiramate showed no significant change in exposure to eslicarbazepine but an 18% decrease in exposure to topiramate, most likely caused by a reduced bioavailability of topiramate. No dose adjustment is required.

Valproate and levetiracetam
A population pharmacokinetics analysis of phase III studies in epileptic adult patients indicated that concomitant administration with valproate or levetiracetam did not affect the exposure to eslicarbazepine but this has not been verified by conventional interaction studies.

Oxcarbazepine
Concomitant use of eslicarbazepine acetate with oxcarbazepine is not recommended because this may cause overexposure to the active metabolites.

<u>Other medicinal products</u>

Oral contraceptives
Administration of eslicarbazepine acetate 1,200 mg once daily to female subjects using a combined oral contraceptive showed an average decrease of 37% and 42% in systemic exposure to levonorgestrel and ethinylestradiol, respectively, most likely caused by an induction of CYP3A4. Therefore, women of childbearing potential must use adequate contraception during treatment with eslicarbazepine Acetate tablets, and up to the end of the current menstruation cycle after the treatment has been discontinued.

Simvastatin
A study in healthy subjects showed an average decrease of 50% in systemic exposure to simvastatin when co-administered with eslicarbazepine acetate 800 mg once daily, most likely caused by an induction of CYP3A4. An increase of the simvastatin dose may be required when used concomitantly with eslicarbazepine acetate.

Rosuvastatin
There was an average decrease of 36-39% in systemic exposure in healthy subjects when co-administered with eslicarbazepine acetate 1,200 mg once daily. The mechanism for this reduction is unknown, but could be due to interference of transporter activity for rosuvastatin alone or in combination with induction of its metabolism. Since the relationship between exposure and drug activity is unclear, the monitoring of response to therapy (e.g., cholesterol levels) is recommended.

Warfarin
Co-administration of eslicarbazepine acetate 1,200 mg once daily with warfarin showed a small (23%), but statistically significant decrease in exposure to S-warfarin. There was no effect on the R- warfarin pharmacokinetics or on coagulation. However, due to inter-individual variability in the interaction, special attention on monitoring of INR should be performed the first weeks after initiation or ending concomitant treatment of warfarin and eslicarbazepine acetate.

Digoxin
A study in healthy subjects showed no effect of eslicarbazepine acetate 1,200 mg once daily on digoxin pharmacokinetics, suggesting that eslicarbazepine acetate has no effect on the transporter P- glycoprotein.

Monoamino Oxidase Inhibitors (MAOIs)
Based on a structural relationship of eslicarbazepine acetate to tricyclic antidepressants, an interaction between eslicarbazepine acetate and MAOIs is theoretically possible.

3.6 Fertility, pregnancy and lactation

Pregnancy

Risk related to epilepsy and antiepileptic medicinal products in general

It has been shown that in the offspring of women with epilepsy using an antiepileptic treatment, the prevalence of malformations is two to three times greater than the rate of approximately 3% in the general population. Most frequently reported are cleft lip, cardiovascular malformations and neural tube defects. Specialist medical advice regarding the potential risk to a foetus caused by both seizures and antiepileptic treatment should be given to all women of child-bearing potential taking antiepileptic treatment, and especially to women planning pregnancy and women who are pregnant. Sudden discontinuation of antiepileptic drug (AED) therapy should be avoided as this may lead to seizures that could have serious consequences for the woman and the unborn child. Monotherapy is preferred for treating epilepsy in pregnancy whenever possible because therapy with multiple AEDs could be associated with a higher risk of congenital malformations than monotherapy, depending on the associated AEDs.

Neurodevelopmental disorders in children of mothers with epilepsy using an antiepileptic treatment has been observed. There is no data available for eslicarbazepine acetate on this risk.

Women of childbearing potential/contraception

Women of childbearing potential should use effective contraception during treatment with eslicarbazepine acetate. Eslicarbazepine acetate adversely interacts with oral contraceptives. Therefore, an alternative, effective and safe method of contraception should be used during treatment and up to the end of the current menstrual cycle after treatment has been stopped. Women of childbearing potential should be counselled regarding the use of other effective contraceptive methods. At least one effective method of contraception (such as an intra-uterine device) or two complementary forms of contraception including a barrier method should be used. Individual circumstances should be evaluated in each case, involving the patient in the discussion, when choosing the contraception method.

Risk related to eslicarbazepine acetate

There is limited amount of data from the use of eslicarbazepine acetate in pregnant women. Studies in animals have shown reproductive toxicity (see Fertility section). A risk in humans (including of major congenital malformations, neurodevelopmental disorders and other reproductive toxic effects) is unknown.

Eslicarbazepine acetate should not be used during pregnancy unless the benefit is judged to outweigh the risk following careful consideration of alternative suitable treatment options.

If women receiving eslicarbazepine acetate become pregnant or plan to become pregnant, the use of Zebinix should be carefully re-evaluated. Minimum effective doses should be given, and monotherapy whenever possible should be preferred at least during the first three months of pregnancy. Patients should be counselled regarding the possibility of an increased risk of malformations and given the opportunity to antenatal screening.

Monitoring and prevention
Antiepileptic medicinal products may contribute to folic acid deficiency, a possible contributory cause of foetal abnormality. Folic acid supplementation is recommended before and during pregnancy. As the efficacy of this supplementation is not proven, a specific antenatal diagnosis can be offered even for women with a supplementary treatment of folic acid.

In the newborn child
Bleeding disorders in the newborn caused by antiepileptic medicinal products have been reported. As a precaution, vitamin K1 should be administered as a preventive measure in the last few weeks of pregnancy and to the newborn.

Breast-feeding

It is unknown whether eslicarbazepine acetate is excreted in human milk. Animal studies have shown excretion of eslicarbazepine in breast milk. As a risk to the breast-fed child cannot be excluded breast-feeding should be discontinued during treatment with eslicarbazepine acetate.

Fertility

There are no data on the effects of eslicarbazepine acetate on human fertility. Studies in animals have shown impairment of fertility after treatment with eslicarbazepine acetate.

3.7 Effects on ability to drive and use machines

Eslicarbazepine Acetate tablets have minor to moderate influence on the ability to drive and use machines. Some patients might experience dizziness, somnolence or visual disorders, particularly on initiation of treatment. Therefore, patients should be advised that their physical and/or mental abilities needed for operating machinery or driving may be impaired and they are recommended not to do so until it has been established that their ability to perform such activities is not affected.

3.8 Undesirable effects

Summary of the safety profile
Adverse reactions were usually mild to moderate in intensity and occurred predominantly during the first weeks of treatment with eslicarbazepine acetate.

The risks that have been identified for eslicarbazepine acetate are mainly class-based, dose-dependent undesirable effects. The most common adverse reactions reported, in placebo controlled adjunctive therapy studies with adult epileptic patients and in an active controlled monotherapy study comparing eslicarbazepine acetate with carbamazepine controlled release, were dizziness, somnolence, headache, and nausea. The majority of adverse reactions were reported in <3% of subjects in any treatment group.

Severe cutaneous adverse reactions (SCARS), including Stevens-Johnson syndrome (SJS)/toxic epidermal necrolysis (TEN) and drug reaction with eosinophilia and systemic symptoms (DRESS) have been reported in post-marketing experience with eslicarbazepine acetate treatment.

Tabulated list of adverse reactions
The following convention has been used for the classification of adverse reactions very common (≥1/10), common (≥1/100 to <1/10), uncommon (≥1/1,000 to <1/100) and not known (frequency cannot be estimated from available data). Within each frequency category, adverse reactions are presented in order of decreasing seriousness.

Table 1: Treatment emergent adverse reactions associated with eslicarbazepine:

System Organ Class	**Very common**	**Common**	**Uncommon**	**Not known**
Blood and lymphatic system disorders			Anaemia	Thrombocytopenia, leukopenia

Immune System disorders			Hypersensitivity	
Endocrine disorders			Hypothyroidism	
Metabolism and nutrition disorders		Hyponatraemia, decreased appetite	Electrolyte imbalance, dehydration, hypochloraemia	
Psychiatric disorders		Insomnia	Psychotic disorder, apathy, depression, nervousness, agitation, irritability, attention deficit/ hyperactivity disorder, confusional state, mood swings, crying, psychomotor retardation, anxiety	
Nervous system disorders	Dizziness, somnolence	Headache, disturbance in attention, tremor, ataxia, balance disorder	Coordination abnormal, memory impairment, amnesia, hypersomnia, sedation, aphasia, dysaesthesia, dystonia, lethargy, parosmia, cerebellar syndrome, convulsion, peripheral neuropathy, nystagmus, speech disorder, dysarthria, burning sensation, paraesthesia, migraine	
Eye disorders		Diplopia, vision blurred	Visual impairment, oscillopsia, binocular eye movement disorder, ocular hyperaemia	
Ear and labyrinth disorders		Vertigo	Hypoacusis, tinnitus	
Cardiac disorders			Palpitations, bradycardia	
Vascular disorders			Hypertension (including hypertensive crisis), hypotension, orthostatic hypotension, flushing, peripheral coldness	
Respiratory, thoracic and mediastinal disorders			Epistaxis, chest pain	

Gastrointestinal disorders		Nausea, vomiting, diarrhoea	Constipation, dyspepsia, gastritis, abdominal pain, dry mouth, abdominal discomfort, abdominal distension, gingivitis, melaena, toothache	Pancreatitis
Hepatobiliary disorders			Liver disorder	
Skin and subcutaneous tissue disorders		Rash	Alopecia, dry skin, hyperhidrosis, erythema, skin disorder, pruritus, dermatitis allergic	Toxic epidermal necrolysis, Stevens-Johnson syndrome, drug reaction with eosinophilia and systemic symptoms (DRESS), angioedema, urticaria
Musculoskeletal and connective tissue disorders			Myalgia, bone metabolism disorder, muscular weakness, pain in extremity	
Renal and urinary disorders			Urinary tract infection	
General disorders and administration site conditions		Fatigue, gait disturbance, asthenia	Malaise, chills, oedema peripheral	
Investigations		Weight increased	Blood pressure decreased, weight decreased, blood pressure increased, blood sodium decreased, blood chloride decreased, osteocalcin increased, haematocrit decreased, haemoglobin decreased, transaminases increased	
Injury, poisoning and procedural complications			Drug toxicity, fall, thermal burn	

3.9 Overdose

Symptoms observed after an overdose of eslicarbazepine acetate are primarily associated with central nervous symptoms (e.g. seizures of all types, status epilepticus) and cardiac disorders (e.g. cardiac arrhythmia). There is no known specific antidote. Symptomatic and supportive treatment should be administered as appropriate. Eslicarbazepine acetate metabolites can effectively be cleared by haemodialysis, if necessary.

4. PHARMACOLOGICAL PROPERTIES

4.1 Pharmacodynamic properties

Pharmacotherapeutic group: Antiepileptics, carboxamide derivatives, ATC code: N03AF04

Mechanism of action

The precise mechanisms of action of eslicarbazepine acetate are unknown. However, *in vitro* electrophysiological studies indicate that both eslicarbazepine acetate and its metabolites stabilise the inactivated state of voltage-gated sodium channels, precluding their return to the activated state and thereby preventing repetitive neuronal firing.

Pharmacodynamic effect

Eslicarbazepine acetate and its active metabolites prevented the development of seizures in nonclinical models predictive of anticonvulsant efficacy in man. In humans, the pharmacological activity of eslicarbazepine acetate is primarily exerted through the active metabolite eslicarbazepine.

4.2 Pharmacokinetic properties

Absorption

Eslicarbazepine acetate is extensively converted to eslicarbazepine. Plasma levels of eslicarbazepine acetate usually remain below the limit of quantification, following oral administration. Eslicarbazepine C_{max} is attained at 2 to 3 hours post-dose (t_{max}). Bioavailability may be assumed as high because the amount of metabolites recovered in urine corresponded to more than 90% of an eslicarbazepine acetate dose.

Distribution

The binding of eslicarbazepine to plasma proteins is relatively low (<40%) and independent from concentration. *In vitro* studies have shown that plasma protein binding was not relevantly affected by the presence of warfarin, diazepam, digoxin, phenytoin and tolbutamide. The binding of warfarin, diazepam, digoxin, phenytoin and tolbutamide was not significantly affected by the presence of eslicarbazepine.

Biotransformation

Eslicarbazepine acetate is rapidly and extensively biotransformed to its major active metabolite eslicarbazepine by hydrolytic first-pass metabolism. The steady state plasma concentrations are attained after 4 to 5 days of once daily dosing, consistent with an effective half-life in the order of 20-24 hours. In studies in healthy subjects and epileptic adult patients, the apparent half-life of eslicarbazepine was 10-20 hours and 13-20 hours, respectively. Minor metabolites in plasma are R- licarbazepine and oxcarbazepine, which were shown to be active, and the glucuronic acid conjugates of eslicarbazepine acetate, eslicarbazepine, R-licarbazepine and oxcarbazepine.
Eslicarbazepine acetate does not affect its own metabolism or clearance. Eslicarbazepine is a weak inducer of CYP3A4 and has inhibiting properties with respect to CYP2C19.

Elimination

Eslicarbazepine acetate metabolites are eliminated from the systemic circulation primarily by renal excretion, in the unchanged and glucuronide conjugate forms. In total, eslicarbazepine and its glucuronide correspond to more than 90% of total metabolites excreted in urine, approximately two thirds in the unchanged form and one third as glucuronide conjugate.

Linearity/non-linearity

The pharmacokinetics of eslicarbazepine acetate is linear and dose-proportional in the range 400-1,200 mg both in healthy subjects and patients.

Elderly (over 65 years of age)

The pharmacokinetic profile of eslicarbazepine acetate is unaffected in the elderly patients with creatinine clearance >60 ml/min.

Renal impairment

Eslicarbazepine acetate metabolites are eliminated from the systemic circulation primarily by renal excretion. A study in adult patients with mild to severe renal impairment showed that clearance is dependent on renal function. During treatment with eslicarbazepine dose adjustment is recommended in patients, adults and children above 6 years of age with creatinine clearance <60 ml/min.

In children from 2 to 6 years of age, the use of eslicarbazepine acetate is not recommended. At this age the intrinsic activity of the elimination process has not yet reached maturation.

Haemodialysis removes eslicarbazepine acetate metabolites from plasma.

Hepatic impairment

The pharmacokinetics and metabolism of eslicarbazepine acetate were evaluated in healthy subjects and moderately liver-impaired patients after multiple oral doses. Moderate hepatic impairment did not affect the pharmacokinetics of eslicarbazepine acetate. No dose adjustment is recommended in patients with mild to moderate liver impairment.The pharmacokinetics of eslicarbazepine acetate has not been evaluated in patients with severe hepatic impairment.

Gender

Studies in healthy subjects and patients showed that pharmacokinetics of eslicarbazepine acetate were not affected by gender.

Paediatric population

Similar to adults, eslicarbazepine acetate is extensively converted to eslicarbazepine. Plasma levels of eslicarbazepine acetate usually remain below the limit of quantification, following oral administration. Eslicarbazepine C_{max} is attained at 2 to 3 hours post-dose (t_{max}). Body weight was shown to have an effect on volume of distribution and clearance. Furthermore, a role of age independently of weight with regards to clearance of eslicarbazepine acetate could not be excluded, in particular for the youngest age group (2-6 years).

Children aged 6 years and below

Population pharmacokinetics indicate that in the subgroup of children aged from 2 to 6 years, doses of 27.5 mg/kg/day and 40 mg/kg/day are required in order to achieve exposures that are equivalent to the therapeutic doses of 20 and 30 mg/kg/day in children above 6 years of age.

<u>Children above 6 years of age</u>

Population pharmacokinetics indicate that comparable eslicarbazepine exposure is observed between 20 and 30 mg/kg/day in children above 6 years old and adults with 800 and 1200 mg of eslicarbazepine acetate once-daily, respectively.

5. SHELF-LIFE

36 months from the date of manufacturing.

6. HOW SUPPLIED/PRESENTATION

Esclicarbazepine Acetate tablets 200mg/400mg/600mg & 800mg: Alu-Alu blister of 1x10, 3x10 and 10x10 tablets with printed packaging insert.

7. STORAGE

Store below 30°C.

8. DATE OF REVISION OF THE TEXT

18 April 2025

9. MANUFACTURED BY:

TORRENT PHARMACEUTICALS LTD.
Bharuch- 392130, INDIA.